Sami Bekuma

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PROFESSIONAL SUMMARY

Machine Learning Engineer experienced in architecting and scaling production LLM, Agentic AI, and NLP systems serving 20M+ daily user interactions. Specialized in multi-agent RAG diagnostics, LLM-as-a-Judge evaluation frameworks, and latency-aware intent routing using LangGraph, Azure OpenAI, AWS Bedrock, and fine-tuned open-source models (Qwen, Gemma). Proven track record of slashing root-cause debugging time by 98%, accelerating CI/CD release validation by 85%, and optimizing GPU inference costs at enterprise scale.

PROFESSIONAL EXPERIENCE

Machine learning Engineer

Comcast - Washington, DC 2024 –present

- Architected and built from scratch an agentic AI failure-investigation platform for Comcast's US/UK production Voice RAG system, combining deterministic triage with LLM reasoning and diagnostic tool selection across policy screening, entity/facet extraction, retrieval, and response generation; reduced root-cause investigation time from 30+ minutes to under 30 seconds across 10K+ triaged queries daily.
- Developed and scaled production NLP services for the Xfinity Voice Remote AI platform, processing 20M+ daily voice commands across North America and Europe and powering multilingual device control and content-discovery experiences, including Olympics and NFL content on Peacock.
- Designed and productionized a supervised intent-classification model using Python, Scikit-learn, and Transformers to intelligently route requests between LLM/RAG pipelines and traditional CPU-based services, reducing infrastructure costs while maintaining response quality.
- Built an automated LLM evaluation framework in Python and integrated it into Azure CI/CD, using Azure OpenAI and a fine-tuned Qwen model for LLM-as-a-Judge evaluation of factuality, hallucination, relevance, toxicity, and multi-turn conversational quality; reduced release-validation cycles by 85%.
- Built an end-to-end AI evaluation platform with Python backend services and a React frontend, enabling engineers to execute evaluation suites, compare model performance, visualize evaluation metrics, and interactively investigate LLM failures.
- Designed and deployed latency-aware load balancing for production voice services, using real-time Grafana observability to monitor latency, throughput, and service health; reduced request timeouts by 9%.

Lead-tutor

Tutoring Club of McLean, *McLean, VA* 2022 – 2024

- Mentored students in Python and Java programming, data structures, and algorithmic problem-solving through structured technical curricula
- Developed structured lesson plans and supported SAT/ACT preparation, improving students' technical confidence and exam readiness.

SKILLS

- **Languages & Backend:** Python (Advanced), SQL, JavaScript, Java, FastAPI, Pydantic, React, Spring Boot, REST APIs, Agent-to-Agent (A2A)
- **AI, LLMs & Machine Learning:** LangGraph, LangChain, LlamaIndex, Transformers (Hugging Face), vLLM, KServe, Scikit-learn, SentenceTransformers, Pandas, NumPy, SpaCy, NLTK, RAG Systems, Fine-Tuning (Qwen, Gemma), Azure OpenAI, AWS Bedrock
- **LLMOps, Evaluation & Cloud Infrastructure:** LLM-as-a-Judge, Model Evaluation & Benchmarking, DeepEval, Langfuse, OpenTelemetry, MLflow, Grafana, AWS S3, Azure CI/CD, PostgreSQL, Docker, Kubernetes

TECHNICAL PROJECTS

Agentic AI Failure Investigation Platform for Voice RAG

Python | LangGraph | LangChain | FastAPI | Agent-to-Agent (A2A) | Pydantic | Azure OpenAI | Gemma | DeepEval | Langfuse | OpenTelemetry | AWS S3 | PostgreSQL

- **Designed a dual-mode diagnostic engine** supporting both live requests and historical S3 production traces, reconstructing and normalizing pipeline evidence for automated root-cause analysis.
- **Built hybrid deterministic triage and LLM-driven workflows** with LangGraph and LangChain, enabling agents to dynamically select diagnostic tools and isolate failures across policy screening, entity/facet extraction, retrieval, and response generation.

- **Implemented cost-aware model routing** across Azure OpenAI and open-source Gemma models, directing targeted DeepEval evaluation workloads to lower-cost models and reducing evaluation token costs by ~45% while preserving frontier models for complex multi-step reasoning.
- **Developed a fault-tolerant FastAPI service** using Agent-to-Agent (A2A) and Pydantic, generating schema-validated diagnostic reports (root cause, supporting evidence, confidence, completeness, and limitations) backed by AWS S3 trace reconstruction and PostgreSQL checkpointing.
- **Integrated Langfuse and OpenTelemetry** for end-to-end observability, tracing multi-step agent trajectories and individual tool calls to monitor latency, token usage, cost, and performance bottlenecks.

AI-Powered Doctor Appointment Booking System (React, Node.js, Express, MySQL, Google Gemini)

- Designed and implemented a **production-style healthcare scheduling platform** that replaces manual and phone-based appointment booking with a structured, self-service workflow.
- Integrated an **AI-powered chatbot (Google Gemini)** that interacts with backend APIs and the database to **create and cancel appointments in real time**, in addition to navigation and user assistance.
- Designed **backend scheduling and conflict-prevention logic** using Node.js, Express, and relational data modeling in MySQL.

Automated Real-Time Data Ingestion Pipeline with AWS and OpenSearch

- Designed and deployed an end-to-end cloud data ingestion pipeline using **AWS** services (**S3**, **SQS**, **Lambda**, **OpenSearch Service**) to automate real-time data processing and analytics.
- Configured an **event-driven architecture** to trigger processing on new **S3** uploads, enabling immediate parsing and indexing into **OpenSearch** for scalable search and visualization.

EDUCATION

GEORGE MASON UNIVERSITY, The Volgenau School of Engineering, Fairfax, VA Dec 2023
BS in Computer Science Grade: 3.64
Honors & Awards: 2022 All-Virginia Academic Team Award, Dean's List, Presidential Scholar